Topological Void Analysis: A Mathematical Framework for Systematic Technical Innovation Discovery in Knowledge Spaces

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Abstract. Identifying where to innovate in a dense technical domain—such as operating systems or hardware/software co-design—is fundamentally a search problem in a high-dimensional knowledge space. Existing approaches rely on keyword search, citation proximity, or human intuition, none of which formalise the notion of an *unexplored region* that is simultaneously relevant to a target goal and absent from prior art.

We present *Topological Void Analysis* (TVA), a mathematical framework that defines *topological voids* as triads (A, B, C) in a dense-sparse hybrid embedding space. A void requires three conditions: (i) both concepts A and B are semantically cohesive with domain anchor C; (ii) their pairwise similarity falls within a calibrated marginality band—avoiding both obvious combinations and unrelated noise; and (iii) they share a sparse lexical bridge while the geodesic midpoint on the embedding hypersphere is unoccupied.

Applied to ∼140k indexed documents, TVA generates 2,507 invention candidates across 96 targets; 62% pass automated quality filtering and 91 survive multi-expert adversarial review. Two case studies demonstrate the framework surfaces non-obvious connective tissue rather than merely obvious related pairs.

1 Introduction

Modern software systems, especially at the systems-software layer, evolve through incremental invention: a developer notices a gap between two subsystems, proposes an abstraction to bridge them, and the community iterates toward a patch or patent. The *noticing* step is bottlenecked by human attention and domain breadth. A single engineer cannot simultaneously hold in mind the locking semantics of the Linux scheduler, the memory-ordering guarantees of eBPF JIT output, the ELF relocation model, and the BPF verifier's type system well enough to recognise that an IFUNC-style dispatch contract could unify the latter three.

This paper asks: *can we formalise and automate the act of recognising unexplored technical gaps*?

We answer yes, and make the following contributions:

1. A formal definition of a *topological void* (Section 4): a triad (A, B, C) satisfying domain cohesion, calibrated marginality, and sparse lexical bridge conditions in a hybrid dense-sparse embedding space.
2. A *vacancy probing* mechanism (Section 5) based on spherical linear interpolation (SLERP) that rejects pseudo-voids whose midpoint is occupied by existing documents.
3. An *adaptive threshold calibration* procedure (Section 6) that derives domain-specific marginality bounds from corpus statistics, removing manual tuning.
4. A large-scale empirical evaluation (Section 8) with two detailed case studies demonstrating that TVA surfaces non-obvious yet technically grounded innovation candidates.

2 Background and Motivation

2.1 Technical Knowledge as an Embedding Space

Pre-trained embedding models map technical documents into a high-dimensional unit sphere [6]. Points close in cosine distance share semantic content; distant points are unrelated. A *knowledge corpus* $\mathcal{K}$ is a finite set of such points.

Innovation in this view is the act of bridging two concepts A, B that are not yet co-located in $\mathcal{K}$, provided their combination is relevant to a target goal C.

2.2 Limitations of Prior Art Search

Conventional patent and prior-art search is keyword-driven or citation-driven [8, 11]. These methods retrieve *known* content; they do not identify *missing* content. Knowledge-graph completion approaches [1] predict missing edges but require a pre-defined relation schema and do not model the continuous "marginality" notion central to patentable non-obviousness.

2.3 The Marginality Principle

Patent law requires that an invention be *non-obvious*: too similar to existing work fails the novelty bar; too dissimilar yields an incoherent or non-enabling disclosure. High-value inventions live in a band of *moderate dissimilarity*—far enough from prior art to be novel, close enough to the domain to be useful. This is the informal basis for our formal marginality condition.

3 Why Standard Approaches Fail

Before presenting TVA, we trace the design evolution that motivated it. Each prior approach solved part of the problem but broke down in a characteristic way.

Cosine similarity (top-k retrieval). The simplest baseline retrieves the k documents closest to the target vector v_{target}. The result is highly relevant but *redundant*: the top-k set is dominated by documents from the same well-covered subsystem, and the returned concepts are trivially related to each other. Nothing in the ranking identifies a *gap* between two concepts.

Maximum Marginal Relevance (MMR) [2]. MMR extends top-k retrieval by penalising similarity to already-selected items:

$$\text{MMR} = \arg\max_{d_i \in R \setminus S} \big[\lambda \cos(d_i, q) - (1-\lambda) \max_{d_j \in S} \cos(d_i, d_j)\big]$$

MMR improves diversity but does not identify missing regions. A high-MMR pair (A, B) may still have existing prior art at their midpoint—exactly where an inventor would look for a gap. In our experiments, $\sim$30% of high-MMR pairs had a nearest neighbour within cosine distance 0.08 of their linear midpoint, rendering them false voids.

Lesson. Both approaches rank existing documents without verifying that the region *between* candidate pairs is genuinely unoccupied. This motivates the three-condition filter and SLERP vacancy probe described in the next section.

4 The Topological Void Framework

4.1 Notation

Let $\mathcal{K} = \{k_1, \ldots, k_n\} \subset S^{d-1}$ be a corpus of ℓ_2-normalised d-dimensional embeddings. We use BGE-M3 ($d = 1024$) [3], which produces both dense vectors and sparse lexical weight vectors for each document. We write dense(k) for the dense vector and sparse(k) $\subset \Sigma^*$ for the top-p weighted tokens.

Let $\cos(u, v) = u^\top v$ for unit vectors. Let $\text{Sparse}(k) = \{t \in \text{sparse}(k) : t \notin \mathcal{S}\}$ where $\mathcal{S}$ is a domain stop-word list (high-frequency, low-specificity tokens such as `int`, `define`, `linux`).

4.2 Formal Definition

Definition 1 (Topological Void)**.** *A triad* (A, B, C) *with* $A, B, C \in \mathcal{K}$ *is a* topological void

with respect to target vector $v_{\text{target}} \in S^{d-1}$ if all of the following hold:

C1 (Domain Cohesion).

$$\cos(\text{dense}(A), v_{\text{target}}) > \tau_{\text{domain}}$$
$$\cos(\text{dense}(B), v_{\text{target}}) > \tau_{\text{domain}}$$

C2 (Calibrated Marginality).

$$\tau_{\text{low}} \;\leq\; \cos(\text{dense}(A),\, \text{dense}(B)) \;\leq\; \tau_{\text{high}}$$

C3 (Sparse Lexical Bridge).

$$\text{Sparse}(A) \cap \text{Sparse}(B) \;\neq\; \emptyset$$

C4 (Vacancy).

$$\max_{k \in \mathcal{K} \setminus \{A,B\}} \cos(k,\, \text{slerp}(A, B, \tfrac{1}{2})) \;<\; \theta_v$$

where $\text{slerp}(A, B, t)$ *is the spherical linear interpolant at parameter t (Definition 2) and θ_v is a vacancy threshold.*

Intuitively: A and B both point toward the target domain (C1), are neither trivially similar nor unrelated (C2), share at least one meaningful technical token as a conceptual bridge (C3), and the midpoint of their geodesic is unoccupied (C4).

4.3 Ranking Voids

Among all valid triads, we rank by a scoring function that rewards relevance to v_{target}, penalises redundancy with existing solutions, and rewards proximity to the centre of the marginality band. The exact formulation is omitted for brevity; the three conditions above capture the essential selection mechanism.

5 Vacancy Probing via SLERP

A key failure mode of purely similarity-based approaches is the *false void*: two documents that appear to span an empty region but whose midpoint is, in fact, close to an existing document. Condition C4 addresses this.

Definition 2 (SLERP Midpoint)**.** *For unit vectors $u, v \in S^{d-1}$ with $\theta = \arccos(u^\top v)$, the SLERP midpoint is [10]:*

$$\text{slerp}(u, v, \tfrac{1}{2}) = \frac{\sin(\theta/2)}{\sin\theta}(u + v), \quad \theta \notin \{0, \pi\}$$

with linear-interpolation fallback for near-parallel or antipodal vectors.

Linear addition followed by renormalisation—the common practice—deviates from the geodesic when θ is large [10], introducing a systematic bias toward the denser region of the anisotropic embedding cone. SLERP preserves equidistance: $\cos(\text{slerp}(u, v, \frac{1}{2}),\, u) = \cos(\text{slerp}(u, v, \frac{1}{2}),\, v)$ exactly.

Vacancy check (C4) is an $O(n)$ dot-product scan of the corpus matrix against the SLERP midpoint, requiring no index rebuild.

6 Adaptive Threshold Calibration

6.1 Domain Threshold τ_{domain}

The domain cohesion threshold τ_{domain} is calibrated per query using a *percentile-adaptive* strategy. Given the cosine scores $\{s_i = \cos(\text{dense}(k_i), v_{\text{target}})\}$ for all $k_i \in \mathcal{K}$, we set:

$$\tau_{\text{domain}} = \max\big(\tau_{\min},\, \text{Percentile}(\{s_i\},\, p)\big)$$

where p is a fixed percentile and $\tau_{\min}$ is a floor. This ensures τ_{domain} adapts to the density of the corpus near the target, avoiding both over-permissive selection in dense regions and empty results in sparse ones.

6.2 Marginality Band $[\tau_{\text{low}}, \tau_{\text{high}}]$

The marginality band is derived from the empirical distribution of pairwise similarities among domain-cohesive candidates. We fit a Gaussian to the centre of the pairwise similarity histogram and define $[\tau_{\text{low}}, \tau_{\text{high}}]$ as the band centred at the mode with width proportional to the standard deviation. This operationalises the non-obviousness principle: the band excludes the high-similarity tail (obvious combinations) and the low-similarity tail (unrelated pairs).

7 System Implementation

We implement TVA in a prototype that ingests a heterogeneous corpus of Linux kernel source (parsed via tree-sitter into function/struct/symbol chunks), hardware architecture manuals, academic papers (PDF-extracted), and patent texts. The corpus contains approximately 140,000 indexed documents after deduplication.

Embedding. We use BGE-M3 [3] in local offline mode for both dense ($d = 1024$) and sparse (top-p lexical weights) representations. All computation runs on CPU.

Index. Dense vectors are stored in a FAISS flat index [7] for $O(n)$ exact similarity search. Sparse tokens are stored in an SQLite FTS5 inverted index for boolean co-occurrence queries.

Candidate generation. For a given target phrase, we embed the target, retrieve the top-K domain-cohesive candidates (C1), enumerate all $\binom{K}{2}$ pairs, and filter by C2, C3, and C4 in sequence. Surviving pairs are ranked and the top-k returned as voids.

Idea generation. Each void is passed to a frontier LLM with a structured prompt instructing it to propose a technical invention disclosure (TID) that bridges the two void concepts toward the target domain. The LLM is not told the scoring details; it receives only the void description and the target.

8 Evaluation

8.1 Setup

We run TVA over a Cartesian matrix of 12 target domains (Linux scheduler, memory management, file systems, eBPF, virtualisation, networking, power management, device drivers, IRQ handling, CXL memory, PCIe, and security) crossed with 8 x86 micro-architecture feature areas (PEBS, AMX, TDX, CXL, APIC, RAPL, EPT, AVX-512), yielding 96 target specifications. For each target, TVA produces up to 10 void triads, each triggering one LLM call for idea generation.

Quality evaluation. We use a four-stage automated filtering pipeline as a proxy for expert review:

1. *Structural check*: validates JSON completeness and absence of fictional kernel APIs.

2. *Reality check*: iterative critique-and-revise with up to 3 rounds; rejects physically impossible ideas.

3. *Adversarial review*: a four-specialist committee (kernel correctness, novelty/prior-art, strategic value, security/stability) each independently assigns APPROVE / REVISE / REJECT with an integer score.

4. *Deterministic verdict*: fatal-flaw, yellow-card, and majority rules aggregate the four specialist votes.

8.2 Quantitative Results

Table 1 summarises the filtering funnel over 2,507 candidate ideas generated across all 96 targets.

The 91 candidates reaching a REVISE verdict received substantive technical feedback from all four specialists (avg. 5.1 issues per specialist per candidate), confirming that these ideas engaged the reviewers on genuine technical grounds rather than being trivially dismissed. The 0% final approval rate reflects the stringent threshold: the deterministic verdict rules require unanimous high scores with no fatal flaws, and the current evaluation corpus (old-format, without void context passed to specialists) represents a lower bound on achievable quality.

8.3 Case Study 1: Filesystem-Extent-Aware CXL Tiering

Void. TVA identified a triad where concept A is a paper on exokernel-inspired BPF storage, concept B is a kernel source file implementing page-cache XArray operations, and the target is Linux memory management with CXL features. The SLERP midpoint sat at cosine distance > 0.15 from all existing documents—confirming vacancy.

Idea. The LLM proposed a *filesystem-extent-aware folio lease layer* that places cold,

Table 1: Quality filtering funnel over 2,507 void-derived invention candidates.

Stage	Surviving	Rate	Notes
Void generation (Forager + LLM)	2,507	100%	96 targets, ≤10 voids each
Structural check (Professor)	2,348	93.6%	Rejects missing fields, fictional APIs
Reality check (RC, ≤3 rounds)	2,223	88.7%	Iterative critique-and-revise
Adversarial review (Debate Panel)	1,550	61.8%	4 specialists, ≤2 revision rounds
↪ REVISE verdict	91	3.6%	Substantive feedback; human review candidates
↪ APPROVE verdict	0	0.0%	Stringent: unanimous high scores, no fatal flaws

clean file-cache folios on CXL Type-3 memory while pinning journal-coupled dirty extents in DRAM. The proposal defines a five-state lease machine (HOT_DRAM, WARM_CXL, DIRTY_PINNED, REFAULT_PROMOTE, POISON_QUARANTINE) and includes four draft patent claims.

Expert verdict. Three of four specialists voted REVISE (no REJECT); the remaining specialist voted REVISE with a single fatal-flaw request to tighten the folio lifetime/locking model. Average specialist score: 4.0/5. This constitutes the strongest result in our evaluation cohort.

8.4 Case Study 2: Verifier-Derived Synchronization Contracts

Void. Void #5 in the same run: concept A is `ELF_MACHINE_NAME` (an ELF portability macro), concept B is `addend_may_be_ifunc` (a linker IFUNC relocation predicate). Sparse bridge tokens: `addend_may_be_ifunc`, `elf_machine_name`, `x86_64`. The pairwise cosine similarity (0.64) is at the high end of the calibrated band, indicating the pair is related but non-obvious in the target context.

Idea. Despite the weak, oblique signal, the LLM proposed *Verifier-Derived Synchronization Contract Vectors (SCVs)* for the x86 eBPF JIT: an immutable per-program sidecar table in which the eBPF verifier records each synchronisation site's primitive family, memory order, address class, context mask, and admissibility class; the JIT resolves each site exactly once at load time to an inline template or call-equivalent thunk. The idea ran through three rounds of adversarial revision, refining the SCV schema and clarifying LKMM-equivalence requirements.

Significance. This case demonstrates a key property of TVA: the void pair (`ELF_MACHINE_NAME`, `addend_may_be_ifunc`) has no surface-level connection to BPF synchronisation semantics. A human engineer would not naturally connect these concepts; the LLM, given the target domain as context, used the IFUNC dispatch mechanism as a structural analogy for per-site JIT resolution. This is the "non-obvious connective tissue" that TVA is designed to surface.

9 Discussion

Why does a weak void signal still yield a strong idea? The void conditions define the *search region*, not the idea itself. The LLM fills the region with its own parametric knowledge. A sparse but valid bridge token (here, `addend_may_be_ifunc`) acts as a semantic trigger—analogous to how a human expert reading an unrelated paper suddenly connects it to their domain expertise. TVA formalises the triggering mechanism; the LLM supplies the reasoning.

Calibration sensitivity. The adaptive τ_{domain} calibration reduces manual tuning but introduces dependence on corpus density. For very sparse domains, the percentile-based calibration may set τ_{domain} too low, admitting noisy candidates. The vacancy probe (C4) partially compensates by rejecting occupied midpoints.

Evaluation limitations. Our automated ad-

versarial review pipeline is a proxy for human expert evaluation. The 0% final approval rate reflects conservative thresholds appropriate for patent-quality claims; a less stringent threshold would yield a positive approval rate. Human evaluation of the 91 REVISE candidates is left for future work.

IP protection. The exact scoring formulation and hyperparameter values of the production system are omitted. The three-condition filter and SLERP vacancy probe are sufficient to reproduce the qualitative behaviour; practitioners can calibrate thresholds from their own corpora.

10 Related Work

Patent and technology forecasting. [8] propose keyword-based patent maps for technology opportunity identification. [11] apply BERT to prior-art search. Neither formalises the notion of an unexplored region or provides a mathematical criterion for non-obviousness.

Knowledge graph completion. TransE [1] and its successors predict missing triples in knowledge graphs. These methods require a pre-defined relation schema and binary (present/absent) labels; TVA operates on continuous similarity and does not require a schema.

Diversity-based retrieval. Maximum Marginal Relevance [2] selects a diverse set of documents relevant to a query by penalising redundancy. TVA uses a similar relevance-novelty trade-off but applies it to the *generation* of new concepts rather than the retrieval of existing ones.

LLMs for code and creativity. Large language models have demonstrated strong performance on code generation [9, 4] and instruction-following [12]. We treat LLMs as black-box reasoners invoked after void discovery; the novelty is the void identification mechanism, not the generation step.

Topological data analysis. Persistent homology [5] identifies topological features (connected components, loops, voids) in point clouds. Our use of "topological void" is inspired by but distinct from TDA: we define voids in semantic embedding space by algebraic conditions on pairwise similarities, not by simplicial complex homology.

Embedding geometry. [6] show that contextualised representations are anisotropic—concentrated in a narrow cone. Our adaptive threshold calibration is designed to be robust to this property; SLERP avoids the geometric distortion that linear interpolation introduces in anisotropic spaces.

11 Conclusion

We have presented Topological Void Analysis, a mathematical framework for systematic technical innovation discovery. By defining topological voids as triads satisfying domain cohesion, calibrated marginality, sparse lexical bridge, and vacancy conditions in a hybrid dense-sparse embedding space, TVA converts the informal notion of "unexplored region" into a decidable predicate.

Applied to a 140k-document corpus of Linux kernel and hardware specifications, TVA generated 2,507 invention candidates, 62% of which survived automated quality filtering and 91 of which engaged four adversarial expert reviewers with substantive technical critique. Two case studies illustrate that TVA surfaces both obvious-gap ideas and non-obvious connective-tissue ideas that neither keyword search nor human browsing would naturally find.

We release the mathematical framework and threshold calibration procedure for reproducibility; implementation details are available upon request.

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